

Exer

Q₁. Derive the Fourier Series of $f(x) = k$, where k is a constant, the periodicity being 2π .

Q₂. Expand in Fourier series of periodicity 2π of
 $f(x) = x \sin x$, for $0 < x < 2\pi$

Q₃. What is the Fourier expansion of the periodic function, whose definition in one period is

$$f(x) = \begin{cases} 0, & \text{when } -\pi < x < 0 \\ \sin x, & \text{when } 0 < x < \pi \end{cases} ?$$

Evaluate $\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \frac{1}{7 \cdot 9} + \dots$ to ∞

Q₄. Obtain the Fourier series of the periodic function defined by

$$f(x) = -\pi, \text{ if } -\pi < x < 0 \text{ and}$$

$$f(x) = x \text{ if } 0 < x < \pi. \text{ Deduce that}$$

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \text{ to } \infty = \frac{\pi^2}{8}$$

Q5. obtain the Fourier series of periodicity 2π for $f(x) = e^{-x}$ in the interval $0 < x < 2\pi$

Hence deduce the value of

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{1+n^2}$$

Further, derive a series for $\operatorname{cosech} \pi$.

Q6. If $f(x) = \begin{cases} \sin x & \text{in } 0 \leq x \leq \pi \\ 0, & \text{in } \pi \leq x \leq 2\pi \end{cases}$

find a Fourier series of periodicity 2π and hence evaluate

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots \text{ to } \infty$$

Q7. Find sine half-range series for the function $f(x) = x$ in $0 < x \leq \frac{\pi}{2}$ and $f(x) = \pi - x$ in $\frac{1}{2}\pi < x < \pi$

Q8. Expand the function $f(x) = \sin x$ $0 < x < \pi$ in Fourier Cosine series.

Q9. Express $f(x) = x(\pi - x)$, as a Fourier Series of periodicity 2π containing
a) Sine terms only
b) cosine terms only.

Hence deduce $1 - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \dots = \frac{\pi^2}{32}$

and $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$